

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1866

Knuaf 100 kV
March 26, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1866. The Interconnection Customer proposes a 149.5 MW solar photovoltaic facility interconnecting at the Knuaf 100 kV substation in the Ohio area. The request was received on August 24, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 539 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$109,120,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Cave Spring / Seton Lake 100 KV	Myrtle Pt / Post ock 230 KV	132.1	92	26.3
Acme Hill / W Salem 100 KV L	Trego / Beaver 500 KV	131.8	127	5.1
Acme Hill / Seville 100 KV	Trego / Bever 500 KV L	130.1	122	5.1
Seville / Ryan 100 KV	Trego / Beaver 500 KV L	127.5	125	5.1
S Hollywood / Kerman 100 KV	Pot Rock / Myrtle Pt 230 KV	127.2	128	29.9
Callender / Jemison 100 KV	Post Rock / Myrtle Pt 230 KV	123.4	116	8.8
Knuaf / Wappingers Falls 100 KV	Post Rock / Myrtle Pt 230 KV	120.1	119	26.3
Wappingers Falls / Seton Lake 100 KV	Post Rock / Myrtle Pt Substation 230 KV	118.7	112	26.3
Quijota / Cave Spring 100 KV	Post Rock / Newport Tie Substation 230 KV	112.3	117	10.4
Kerman / Knuaf 100 KV line	Knuaf / Wappingers Falls 100 KV	111.2	125	100.0
Quijota Substation / Coralville Gt Substation 100 KV	Post Rock / Newport Tie 230 KV Line	110.3	105	10.4
Santa Isabel / Denmark Substation 100 KV	Post Rock / Newport Tie 230 KV	108.6	151	7.2
Mcgill / S Hollywood 100 KV	Post Rock / Myrtle Pt 230 KV	102.6	267	17.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cave Spring / Seton Lake 100 KV	Rebuild line with higher-capacity conductor	\$19,155,000	100% (\$19,155,000)
West Salem / Acme Hill Substation 100 KV Line	Rebuild line with higher-capacity conductor	\$8,869,000	100% (\$8,869,000)
Acme Hill / Seville 100 KV	Rebuild line with higher-capacity conductor	\$2,707,000	100% (\$2,707,000)
Seville / Ryan 100 KV	Rebuild line with higher-capacity conductor	\$3,503,000	100% (\$3,503,000)
S Hollywood / Kerman 100 KV L	Rebuild line with higher-capacity conductor	\$2,315,000	100% (\$2,315,000)
Jemison Substation / Callender 100 KV	Rebuild line with higher-capacity conductor	\$7,140,000	100% (\$7,140,000)
Knuaf / Wappingers Falls 100 KV	Rebuild line with higher-capacity conductor	\$6,229,000	100% (\$6,229,000)
Wappingers Falls / Seton Lake 100 KV	Rebuild line with higher-capacity conductor	\$6,060,000	100% (\$6,060,000)
Quijota / Cave Spring 100 KV	Rebuild line with higher-capacity conductor	\$35,002,000	100% (\$35,002,000)
Knuaf / Kerman 100 KV	Rebuild line with higher-capacity conductor	\$7,426,000	100% (\$7,426,000)
Quijota Substation / Coralville Gt Substation 100 KV	Rebuild line with higher-capacity conductor	\$8,751,000	100% (\$8,751,000)
Santa Isabel / Denmark 100 KV	Reconductor line	\$1,030,000	100% (\$1,030,000)
Mcgill / S Hollywood 100 KV	Reconductor line	\$1,138,000	82% (\$933,000)
	Total allocated to project		\$109,120,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.